Mechanising Count-One Ownership for Array Updates in Lean 4

Gavin Mendel-Gleason
Scidonia Limited

Abstract

A reference-counted array update copies whenever the array is shared or persistent, decided at run time from heap state the source never mentions. We define a count-one ownership discipline for a fragment of Lean 4's compiler IR and mechanise it against model semantics. Its foundation is a complete accounting relation equating each count with the fields that reference the cell plus the owning variables that denote it, which at count one forces one owning root with no edge, or one edge with no root. Preservation is proved for a restricted flat fragment, and separately for the extraction, call and restoration protocol at its accounted boundaries, since the temporary pair the emitted code allocates makes the relation false between them. For the recognised fragment, the counted interpreter's whole run, through restoration and return, is proved to be heap-compatible with the model run and to agree with it in result, fault and trace under an explicit counted child-call contract. The result is a checked count one for that fragment under recognition, layout, entry and call premises: not a physical count, a step-by-step account in the pair-live window, or a claim about C.

2012 ACM Subject Classification Theory of computation → Logic and verification; Software and its engineering → Software verification and validation

Keywords and phrases reference counting, ownership, compiler verification, Lean 4, separation logic

1 Introduction

An array update should cost the same whoever else holds the array. In Lean 4 it does not: arrays are reference counted, and a write mutates in place exactly when the array is uniquely referenced and not persistent, and otherwise copies the whole array [3, 9, 10]. Lean's `lean_is_exclusive` makes that decision at run time, from heap state the source never mentions, so n updates over an n-element array can cost linear or quadratic time according to sharing alone. A static ownership discipline would make the in-place branch a compile-time guarantee; Lean 4 has no ownership or linear types, and adding them would be a language change. This paper recovers the guarantee without one, for a fragment of the compiler's IR, by an external count-one discipline whose soundness is proved in Lean 4 against model semantics.

The first obstruction is that a count of one is not one owner. The natural invariant equates the heap's stored count with the context's multiplicity, but binding a second owned variable leaves that equality true while two variables now denote the cell; a kernel-checked countermodel (`alias_not_accounted`) exhibits exactly that state. The repair is complete accounting: a duplicate-free support enumerating every owning variable, a domain covering every live cell, and the equation $rc(r) = \mathit{IncomingCount}(r) + \mathit{EnvOwnerCount}(r)$, whose right side counts the heap fields that reference r and the owning variables that denote it. At count one the equation admits exactly two shapes—one owning root and no incoming edge, or no owning root and exactly one incoming edge—and two owning variables at one cell are the same variable. The integer now has a meaning.

The second obstruction is the emitted code. A nested update takes the element out of the array, allocates a pair holding parent and child, projects both handles, frees the pair, calls on the child and puts the result back. At the take-out the pair's field duplicates the parent reference without raising the count, while the caller's handle still owns it, so the

© Gavin Mendel-Gleason;
licensed under Creative Commons License CC-BY 4.0

equation reads one = one plus one; it becomes true again at entry, after the release and after restoration, and nothing in between is accounted. The proof is therefore a boundary simulation: count identities for the pair's traffic, restoration's in-place branch derived from the protocol and call boundaries, and a compatibility relation that lets the counted cursor run one ahead of the model's. Closing the fragment needs a modular boundary. The child call is a contract, and its counted version must also supply the caller's accounting at the body's return, which the finite-path induction over model paths does not construct.

We prove, for one level of a finite supported path under that contract, that the counted interpreter's run over the whole recognised fragment—extraction, child check, call, restoration and return—is related to the model's run of the same term by heap compatibility, with the same result, fault and write trace (`principal_counted_fragment`). A second theorem composes recognition, the finite-path model correspondence and the counted boundary accounting (`principal_supported_path`), and a closed kernel-checked instance inhabits it at the recogniser's layout. The closest predecessor, Huisinga's static uniqueness analysis for a Lean IR model [4], enforces uniqueness by a static analysis and leaves soundness and the compiler connection as future work; here the analysis is a proved relation over an existing IR, and the connection to the emitted fragment is checked. The guarantee remains relative to entry, layout, recognition and call premises, and stops at the model heap.

Three contributions.

- **Complete accounting and its preservation.** An owning-variable support, a live-cell domain and the roots-plus-edges equation yield the count-one dichotomy, reject the two-owner countermodel, and make preservation provable: `supported_run` takes an accounted state and one supplied slot fact and concludes a fault-free run of the supported fragment with exclusively flagged writes and accounting at every state reached. The proof says why: a move transfers an owner rather than adding one, a single decrement removes one root together with one count, and a scalar-for-scalar store adds and removes no edge.
- **A counted protocol at its accounted boundaries.** The pair's traffic is neutral at release, the negative controls leave counts elevated, and restoration's mutate branch is derived rather than assumed; `counted_nested_chain` composes take-out, an admitted child call and restoration into one chain, related to the model by a cursor-tolerant compatibility relation. The technique is what survives a correct protocol under which the simple invariant is deliberately false: count identities inside the window, accounting at its boundaries.
- **A checked connection from emitted LCNF to the models.** A recogniser walks the actual declaration, pinning operation order, arities, operand identities and branch constructors, and its trace is executed and related to the model. The flat composed guarantee relates recognised execution, model run and runtime model at the designated in-place write with no accounting premise; the nested statement composes the finite-path correspondence, the counted chain and the counted interpreter's whole-fragment run at one supported level, from an accounted caller state and the counted child-call contract, with a closed instance discharging every premise.

The rest of the paper follows that order: the example and the two languages (Section 2), accounting (Section 3), the checker and flat preservation (Section 4), the counted protocol (Section 5), the compiler connection (Section 6), the instance and evidence (Section 7), and the comparison and limits (Sections 8 and 9).

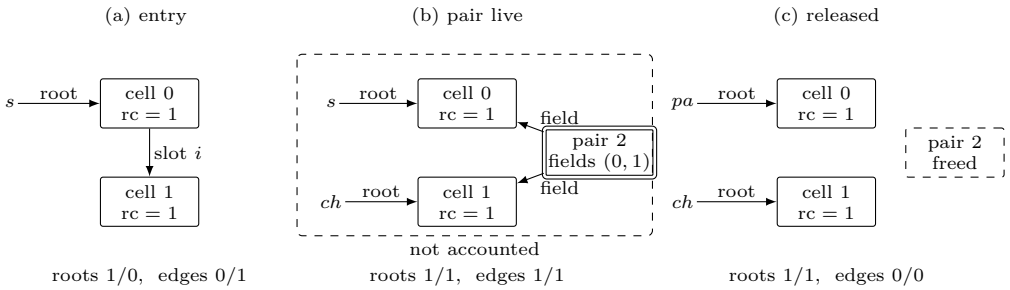


Figure 1 Ownership traffic for one nested update. Each panel is one state of the protocol of Section 5; the pair is the fresh name 2 and i is the selected slot. Counts are per cell, printed as roots over edges for parent and child. The middle panel is outside the accounting relation of Definition 2: the pair's field is an incoming edge for cell 0 while s is still an owning handle for it, so the equation reads $1 = 1 + 1$; the kernel refusal is `pairPhase_refused`. The first and third panels are accounted, at the entry and at the protocol's boundary.

2 The running example and the semantic setting

Lean 4 compiles source through a sequence of IRs. Its last LCNF phase inserts reference-count operations and the backend emits C; `lean_is_exclusive` tests a stored count against one and a persistence flag. The object here is single-threaded, and count one with non-persistence is the whole of the test we model. The development pins the toolchain, the sibling revision and the captured LCNF artifacts, so every claim names the code it is about. The problem is not to implement reference counting but to know statically that the count at a particular write is one; since Lean's source has no linearity, we state the property over the compiler's code and prove the connection, rather than extend the type system.

2.1 The update and its ownership traffic

The source is the array case of `Layout5.setAt` in the sibling: take the element out, check the extracted focus, recurse on it, put the result back.

```
| _, .index i q, y, s =>
    if h : i < s.size then
      match takeOutAt (assertExclusiveArray openSite s) i h (stateDefault _) with
      | ⟨(sib, focus), hbound⟩ =>
        plugAt sib i hbound (setAt q y (assertExclusive structSite focus))
    else s
```

The emitted code is the sequence the development names: a `lean_is_exclusive` test for the array, the take-out exchange (`lean_array_fget`, `lean_array_fset` and the pair allocation), the child check on the extracted element, the recursive call, and the `lean_array_fset` put-back. Figure 1 draws the same three states throughout the paper: the entry, where one root and one slot edge account for the two cells; the take-out, where the temporary pair holds the parent in a field while the caller's handle still owns it; and the release, where the pair is gone and the two projected handles own the cells again.

2.2 Two languages, one accounted subset

The checker does not run on Lean source or on LCNF. Its programs are two small inductive types, interpreted over a model heap and environment; the compiler connection is a mapping

from recognised LCNF to these values.

```
inductive Code where
  | let (x : Var) (v : LetF) (k : Code)
  | inc (x : Var) (n : Nat) (k : Code)
  | dec (x : Var) (n : Nat) (k : Code)
  | check (x : Var) (k : Code)
  | setArr (x : Var) (i : Var) (v : Var) (k : Code)
  | ret (x : Var)

inductive NCode where
  | base (c : Code)
  | takeOut (a i fst snd : Var) (e : Ref) (k : NCode)
  | putBack (a e i result : Var) (k : NCode)
  | callAt (state path x result : Var) (body : NCode) (k : NCode)
  | ret (x : Var)
```

A `Code` is the straight-line array-write fragment; an `NCode` adds extraction, restoration and the recursive call. A `LetF` is one of `lit n`, `size a`, `isShared x` or `checkAlias x`. Variables are `Nat`, references are `Nat`, `Val` is a reference, a scalar or erased, and a `Heap` maps references to objects with a field array, a count and a persistence flag. `run`/`runN` execute these values and `check`/`checkN` are the checkers; every soundness theorem is about this semantics. They are neither Lean source nor LCNF: `projectPrefix` and `projectNested` walk an LCNF value, `Prefix.toModel` and `indexModelTerm` read a `Code`/`NCode` off the recognised operands, and `execAST`/`execNestedAST` execute the recognised trace.

The two languages play different roles and must not be conflated. `Code` and `NCode` are the checker's *general* language: it contains increments, multi-count decrements and arbitrary nesting, and the checker accepts programs that carry no accounting. The theorem-bearing fragment is `Supported` (Section 4.2), an inductively defined subset of `Code` with extra certificates. Checker acceptance, supported certification, state validity and the child-call specification are four separate notions; no result here says that `check c s = some c'` alone preserves accounting.

3 Complete ownership accounting

The model heap, its counts and exclusivity are the semantic side of the judgement: `Inv` ties the context's multiplicity to the heap count, and `Exclusive H r` means that `r` is a live, non-persistent cell whose count is one.

▶ **Definition 1** (The raw invariant). *`Inv H env c` equates the heap's stored count with the context's multiplicity (`rc_eq`); an owning variable denotes its reference at multiplicity at least one (`owned`); a borrowed variable also denotes a live cell (`borrow`); a scalar variable denotes a scalar (`scalar`); and no live cell is persistent (`nonPersistent`).*

`Inv` relates the heap count to the context multiplicity but does not count the environment's owning variables: `Inv.owned` only requires $1 \leq c.mult\ r$. `bindOwned` preserves `Inv` without changing `mult`, so two owned variables can denote one count-one cell, and the raw invariant does not imply physical uniqueness. The kernel countermodel `alias_not_accounted` constructs exactly that state from `initCtx`, which has one owner and does satisfy `OwnersAccounted` (`initCtx_accounted`). `CountFidelity` is not a consequence of `Inv`; it is a consequence of the relation below.

▶ **Definition 2** (Complete accounting) *structure Accounted (support : List Var) (H : Heap) (env : Env) (c : Ctx) : Prop where*

```
  inv              : Inv H env c
  support_nodup    : support.Nodup
  support_sound    : ∀ x, x ∈ support → c.us x = .owned
  support_complete : ∀ x, c.us x = .owned → x ∈ support
  support_bound    : ∀ x, x ∈ support → ∃ r, c.refOf x = some r
  domain           : ∀ r o, H.cells r = some o → r < H.next
  count            : ∀ r, H.rc r = IncomingCount H r + EnvOwnerCount c support r
```

The relation is deliberately redundant, because each field of it is refutable: a duplicate in the support counts a root twice (`dupSupport_witness`), an omitted owner loses one (`omittedOwner_witness`), and an unbound root has no denotation to count (`unboundRoot_witness`). The support is a duplicate-free enumeration of every owning variable, each entry is bound to a reference, every live cell lies inside the range the edge count enumerates, and the count equation holds over two independently computed quantities.

```
def IncomingCount (H : Heap) (r : Ref) : Nat :=
  ((List.range H.next).map fun q =>
    match H.cells q with
    | some o => (o.fields.toList.filter (fun v => v = .ref r)).length
    | none => 0).sum

def EnvOwnerCount (c : Ctx) (vars : List Var) (r : Ref) : Nat :=
  (vars.filter (fun x => c.us x = .owned ∧ c.refOf x = some r)).length

def CountFidelity (H : Heap) (c : Ctx) (vars : List Var) : Prop :=
  ∀ r, H.rc r = IncomingCount H r + EnvOwnerCount c vars r
```

▶ **Lemma 3** (Count-one dichotomy). *If `Accounted support H env c` and $H.rc\ r = 1$, then either $EnvOwnerCount\ c\ support\ r = 1$ and $IncomingCount\ H\ r = 0$, or $EnvOwnerCount\ c\ support\ r = 0$ and $IncomingCount\ H\ r = 1$.*

Proof. Read the count equation through `Inv.rc_eq` (`mult_eq`) and split the two natural numbers whose sum is one. `count_one_dichotomy` is the statement. ◀

▶ **Corollary 4** (Uniqueness at a known root). *If `Accounted support H env c`, $H.rc\ r = 1$, and an owning variable $x \in support$ has $c.refOf\ x =$ some r, then x is the only owning variable at r and r has no incoming edge; `at_most_one_owner` states that any two such variables coincide.*

This is the count-one consequence that gives the integer a meaning, and `initCtx` establishes the relation at the initializer (`initCtx_accounted`). It is a statement about owning variables and heap edges, not about physical pointers: no representation theorem connects a model reference to a runtime object.

4 The checker and flat preservation

4.1 The judgement, once

A context records, for each variable, whether it is owned, borrowed, scalar or dead, which reference it denotes, and the multiplicity of that reference. The checker is a total function from a context and a program to a successor context or failure; a frame stack records open extractions.

Rule	Obligation	Effect
`let y (checkAlias x)`	`x` owned, multiplicity one; `y` not owned	`y` becomes the checked alias; `x` dead
`setArr x i v`	`x` owned, multiplicity one; `i`, `v` scalar	exclusive write through `x`
`dec x 1`	`x` owned; $1 <$ multiplicity	multiplicity -1; `x` dead
`takeOut a i fst snd e`	`a` owned, multiplicity one; `i` scalar; `e` multiplicity one; `fst`, `snd` dead and distinct; no open frame at `a`	push the frame; bind parent and element
`callAt state path x result b`	`state` owned, multiplicity one; `path`, `x` scalar; `result` dead; `CountFree b`; body accepted from the isolated context and returns `state`	call the body; bind the result

Table 1 Selected checker rules. Each premise is an ownership obligation; the successor context is computed, not inferred.

```
inductive U where
  | owned | borrow | scalar | dead

structure Ctx where
  us    : Var → U
  refOf : Var → Option Ref
  mult  : Ref → Nat
```

A frame records the array, index and element of an open extraction; `check : Ctx → Code → Option Ctx` and `checkNF : Ctx → List Frame → NCode → Option Ctx` are total.

The judgement.

The typing judgement $c \vdash s \Rightarrow c'$ is *defined* to hold exactly when `check c s = some c'`: the context is the type of the program point, and there is no separate derivability relation to relate to it. The object language is untyped; there is no type former for a program, and the checker does not infer one by unification. The judgement is syntax-directed—one rule per constructor, the context an input, the successor computed by `setUs`, `move`, `bindOwned`, `inc` and `dec`—so there are no metavariables and the only computation is the context update. Table 1 gives the rules the rest of the paper uses; the remaining constructors (literal, size, sharing, check, put-back and return) have the same shape: a usage-and-multiplicity premise, and a successor computed by those updates, with `ret` refusing a frame it does not close.

Two consequences matter later. `dec` is affine, not the last drop: its premise is $n <$ the object's multiplicity, so it consumes a duplicate handle and leaves the object live. And linearity at the write is the side condition $multOf\ x = 1$, not the usage tag alone.

▶ **Example 5** (A unique write is accepted, an aliased one refused).

```
  def execSet (x y i v : Var) : Code :=
  .let y (.checkAlias x) (.setArr y i v (.ret y))

def aliasSet (x y i v : Var) : Code :=
  .inc x 1 (execSet x y i v)

#guard (check (initCtx 0) (execSet 0 3 1 2)).isSome
#guard (check (initCtx 0) (aliasSet 0 3 1 2)).isNone
```

The alias increments the handle, so the write's multiplicity-one obligation fails; the same refusal appears at the context level for a heap whose count is two. This is the static counterpart of the runtime copy: the checker rejects the program before it can copy.

4.2 The supported fragment

`Supported c s` is indexed at the pre-state `c`; each constructor states the checker's own rule at `c` and its continuation at the checker's post-state, and carries the certificates the checker cannot see: a fresh alias target and a fresh scalar-`.let` target. The fragment has eight constructors; `.inc` is absent and every `.dec` is the single-count form `.dec x 1`, whose premise `1 < c.mult r` is exactly the checker's. From `Supported c s` the checker accepts the program (`supported_check`), which is what lets a preservation theorem follow a fault-free run.

The store's slot fact is deliberately not a constructor premise: no context records which value a cell's selected slot holds, so it is the `StoreSlots` premise of the preservation theorem instead. `StoreSlots H env s` is a recursive predicate over the supported program, supplied from the heap and environment alone (it may not assume the invariant): at the store the run reaches, the array cell's selected slot holds a scalar, so the in-place write replaces a value rather than an edge. `storeSlots_init` discharges it for the initializer's measured write, and nothing general.

4.3 Preservation

▶ **Theorem 6** (Supported execution preserves accounting). *Let `Supported c s`, `Accounted support H env c` and `StoreSlots H env s`. Then there are c', H', env', v and ws with `check c s = some c'` and `run s env H = .done H' env' v ws`, such that every write in ws is flagged exclusive, `Accounted (resultSupport s support) H' env' c'`, and `AccountedAlong H env c support s`.*

```
theorem supported_run {c : Ctx} {s : Code} (hs : Supported c s)
    {support : List Var} {H : Heap} {env : Env}
    (hacc : Accounted support H env c) (hslots : StoreSlots H env s) :
    ∃ c' H' env' v ws,
      check c s = some c' ∧
      run s env H = .done H' env' v ws ∧
      (∀ w ∈ ws, w.2 = true) ∧
      Accounted (resultSupport s support) H' env' c' ∧
      AccountedAlong H env c support s
```

`AccountedAlong` is a conclusion, never a hypothesis: at each instruction the relation holds at the interpreter's own post-state, and at each array store the target is exclusive (`supported_store_exclusive` states the store's own pre- and post-state).

Proof sketch.

Induct on the derivation of `Supported c s`, generalising the support, the heap and the environment; the measure is the syntax of the supported program and the induction hypothesis is applied to the continuation at the state the step produces. Each case is a step of `run` followed by the induction hypothesis, and the content is in three cases. *Fresh binding* (`bindAlias`): the alias target takes over the source's owning obligation, and the support variable is renamed by `replaceVar`; `Accounted.move` shows that `EnvOwnerCount` is unchanged, because the

enumeration follows the handle rather than the name. Freshness is what makes this a transfer: were the target already an owner, the support would gain a duplicate and `support_nodup` would fail. The scalar binders (`bindLit`, `bindSize`, `bindShared`) use `Accounted.setScalar`, which adds no owner and no edge. *Decrement* (`decOne`): the support loses `x`, the heap loses one count, and `IncomingCount` is unchanged, so both sides of the equation at r fall by one (`Accounted.decOne`); the multi-count decrement is excluded precisely because it would release counts that no obligation carried (`decTwo_rejected`). *Edge-neutral scalar store* (`setArr`): the slot fact gives that the selected slot holds a scalar, and the run writes a scalar, so the set of incoming edges is literally unchanged; the equation is preserved by `Accounted.setArrInPlace` at the checker's unchanged context, which is where the write's exclusivity and its trace event are produced. The remaining cases commute with the induction hypothesis.

5 Counted protocol and boundary simulation

5.1 The pair and its boundary

The take-out builds a pair of the parent and the extracted child; the projections increment both components; freeing the unique pair releases both edges. At the boundary the counts return to their entry values only because the pair is freed.

▶ **Theorem 7** (Pair accounting). *Let `WellFormedEntry H p c i`: both components live, non-persistent and at count one, p's selected slot holding c, and the fresh name empty. Let B = protocol H p c i return the fresh name B_1 and the heap B_2. Then*

$$B_2.rc\ p = 1, \quad B_2.rc\ c = 1, \quad B_2.cells\ B_1 = \text{none},$$

both components are `Heap.Exclusive` in B_2, and their counts are H's.

```
theorem pair_accounting {H : Heap} {p c : Ref} {i : Nat}
    (hw : WellFormedEntry H p c i) :
    let B := protocol H p c i
    B.2.rc p = 1 ∧ B.2.rc c = 1 ∧ B.2.cells p ≠ none ∧
      B.2.cells c ≠ none ∧ B.2.cells B.1 = none ∧
      Heap.Exclusive B.2 p ∧ Heap.Exclusive B.2 c ∧
      B.2.rc p = H.rc p ∧ B.2.rc c = H.rc c
```

The no-free control, `no_free_components_not_exclusive`, leaves both components at count two, neither exclusive, and the pair live: the boundary is produced by the free, not by the projections.

5.2 The counted call and restoration

`counted_takeOut` consumes the array's cell and selected slot plus the context facts from `checkNF`; it derives the boundary accounting, `HeapCompat`, the pair's freshness and the cursor advance. The child call is where the modular boundary sits. `CountedChildCall` adds one field to the model child-call contract:

```
structure CountedChildCall (Hin Hout : Heap) (env : Env) (ctx : Ctx)
    (state path x : Var) (body : Nested.NCode) (p c : Ref)
    (support : List Var) : Prop where
  call      : PutBack.ChildCall Hin Hout env ctx state path x body p c
  accounted : Accounted support Hout env ctx
```

Stage	`EnvOwnerCount`	`IncomingCount`	parent rc	child rc	disposition / why
entry	1 / 0	0 / 1	1	1	root handle; child in the parent's slot
take-out	1 / 1	1 / 1	1	1	handles moved; pair fields are the edges
projections	1 / 1	1 / 1	2	2	`incPairComponents_eq`: edge plus handle
free unique pair	1 / 1	0 / 0	1	1	`pair_traffic_neutral`: edges released
child call	1 / 1	0 / 0	1	1	`CountedChildCall`: parent suspended
put-back	1 / 0	0 / 1	1	1	`putBack_exclusive_of_wellFormed`: mutate

Table 2 The counted protocol at an accounted caller entry. Each root/edge pair is printed ordered parent/child. The take-out row is the state of Figure 1(b), where the stored count and the equation disagree: the row's counts are per-stage effects, not boundary facts. The boundary rows are the first, the free and the put-back; `Accounted` is asserted only there.

`call` is the boundary the put-back consumes. `accounted` is the caller's complete accounting at the body's return, and it is an *explicit premise*: the finite-path induction derives the ordinary `ChildContract` at every level (`childContracts_of_path`) but not this field, because the ordinary contract does not mention the caller's ownership state. The closed instance supplies it from the caller's own accounting at the return (`calledAccounted`). From the call's output state, `counted_restore` derives the mutate branch and the accounted post-state, and `putBack_exclusive_of_wellFormed` concludes `Heap.Exclusive` for both the parent and the restored child.

▶ **Theorem 8** (Counted chain). *From an accounted caller entry, take-out, an admitted `CountedChildCall` and restoration compose into one chain: `counted_nested_chain` concludes the final counted state, its accounting, the restoration's `.mutate` disposition, and `HeapCompat` to `runN`'s state; the restored slot's own equation is `counted_nested_chain_restoration`.*

5.3 Compatibility and the boundary argument

▶ **Definition 9** (Heap compatibility). *`HeapCompat Hc Ha` holds when Hc and Ha hold the same cell at every name and $Ha.next \leq Hc.next$. At the protocol instance the slack is exactly the pair allocation the counted side performs and the model's take-out does not; the relation itself permits any slack, and `HeapCompat.eq_of_cursor_eq` shows that with equal cursors the heaps are equal, so it is equality of cells up to the counted side's allocation, not a weakened equivalence.*

The pair-live window is not an accounting failure but a deliberate boundary abstraction, and the case we exhibit is forced: at the take-out state of Figure 1(b), at the context the run's own take-out rule produces, no support satisfies `Accounted` (`pairPhase_refused`), because the count equation reads one = one plus one. The positive control is the boundary, where the counts match again. Three facts make the window safe. The projections' effect is exactly the pair's component increment (`incPairComponents_eq`). The release is boundary-neutral (`pair_traffic_neutral`): the parent's and child's counts are their entry counts, the child's cell is the entry's, and the temporary pair is gone. And the suspended parent

remains exclusive across the child call because the child's contract is confined to the child's mutable footprint rather than because the parent's exclusivity is assumed: `callAt_frame_confined` takes `AnnotationsIn`, checker acceptance, the return variable, the isolated `Inv` and `TakeOutsOK`, and the parent/child separation is a `ChildCall.separated` fact used in the path derivation. Finally the put-back's mutate branch follows from the protocol boundary and the child-call boundary, not from an assumed final state.

Proof sketch.

The protocol argument is a decomposition, not an induction: the five heap-touching operations are discharged in order, each by a count identity on the heap it produces, and the composition is `counted_nested_chain`. Transfer into the pair moves the handles and leaves the pair's fields as the edges; projection adds one root per component; release removes the pair's two edges and nothing else; the call is suspended at the boundary; restoration reads the child's output state and writes the slot. Because each identity is stated at the state it produces, the chain never needs `Accounted` inside the window—only at its ends—and `HeapCompat` carries the one-cell cursor slack that the allocation leaves behind.

6 Recognised-fragment correspondence

6.1 Recognition, and the flat guarantee

A source-level discipline is only useful if it survives lowering. The compiler rewrites programs, inlines helpers and chooses the emitted shape, so the recogniser walks the actual LCNF value, pinning operation order, arities, operand identities and branch constructors. `projectPrefix` accepts the flat write, `projectNested` the `Path.index` arm and `projectPair` the two pair arms; token-preserving near misses are rejected (eleven at the top level, eleven on the nested path, fourteen per pair arm), so the correspondence is about the code we name. No universal rejection theorem is claimed; the claim is scoped to the accepted grammar and the mutation corpus. `project_sound` gives executed/model equality for the abstract prefix, and `execAST_recognized` is decoding adequacy: it relates the trace to the model, it is not a semantics.

▶ **Theorem 10** (Flat principal guarantee). *Let `projectPrefix code = some p`, `Layout assign p`, the entry invariant `EntryInv H env (Runner.initCtx root) root client`, the machine/model agreement `Agrees assign p s env H root i v`, the cell and index facts $H.cells\ root = \text{some}\ o$ and $i < |o.fields|$, and a runtime correspondence `Corr H R` with $R.cells\ root = \text{some}\ o'$. Then the executed prefix's outcome and the model's run both equal `inPlaceOutcome root i v o H`, and for some H' the runtime's `setAt` returns `.inPlace H' root` with `Corr` preserved, the target root and the slot holding the written scalar.*

```
theorem principal_write
    (hproj : projectPrefix code = some p) (hl : Layout assign p)
    (hEntry : EntryInv H env (Runner.initCtx root) root client)
    (hag : Agrees assign p s env H root i v) (hcell : ...) (hin : i < o.fields.
    size)
    (hc : Corr H R) (ho' : R.cells root = some o') :
    PrefixOutcome.ofExec assign p (execAST assign p code s) =
        some (inPlaceOutcome root i v o H) ∧
      PrefixOutcome.ofModel (Ownership.run (Prefix.toModel assign p) env H) =
        inPlaceOutcome root i v o H ∧
      ∃ H', Runtime.setAt R root i (.scalar v) = some (.inPlace H' root) ∧ ...
```

The proof is the conjunction of `recognized_write_in_place` and `Runtime.setAt_inPlace_of_entryInv`, and it takes no accounting premise. `Accounted` is consumed at the initializer by `principal_accounted`, which yields `OwnersAccounted`, `CountFidelity` and the count-one dichotomy. `Prefix.toModel_supported` puts the certified flat model in the counted fragment, so `supported_run` applies once `StoreSlots` is supplied—at `execSet` by `storeSlots_init`, not for a general *p.toModel*. The two results are adjacent, not one theorem.

6.2 The finite path and the counted fragment

The model nests take-out, call and put-back per level. For a finite level path the ordinary child contract is derived at every level: `childContracts_of_path` takes the level list and a supplied path contract whose one function-level ingredient is the leaf computation (agreement, checker acceptance, exit variable, count-freedom, annotation), the rest being the level's slot, environment, context, shape and heap facts, and yields the ordinary `ChildContract` at each level, whose body is the remaining levels' `Path.build`; `index_path_simulates_model` concludes that the executed nested prefix's `NestedOutcome` is the model run of `Path.build path value body (l :: rest)`, completing with an all-exclusive trace under the recogniser's verdict, layout, level match, agreement and path contract. The counted call is not a consequence of that induction (Section 5). Two bounds are recorded rather than hidden: the flat environment cannot express the emitted per-invocation `oproj[1]` index, so a deeper level that re-indexes by re-projecting the continuation is outside the model; and the emitted callee is not captured—`execNested`'s call rule runs the admitted model body with only the call's state, path and value (N10).

▶ **Theorem 11** (Counted fragment correspondence). *Let the recogniser accept the declared shape (`projectNested` yielding p), let the layout, the machine-to-model agreement `NestedAgrees` and the level match hold at one level of a finite path, let the client's path contract and the caller's complete accounting `Accounted support H mEnv c` hold, and let the counted child call be supplied at the protocol's boundary with the caller's accounting at the body's return. Then, writing H_f for the counted final heap:*

1. *`execCountedAST` runs the whole recognised fragment p.fragment from the entry machine to a completed state with heap H_f, result the caller's array handle, restored binder holding that handle, and write trace the entry trace followed by the child's;*
2. *the model run of `Path.build path value body (l :: rest)` completes with that handle and an all-exclusive trace;*
3. *the counted and model outcomes are `NestedOutcome.Compat`: they agree by `HeapCompat` and literally on the result, the fault and the write trace;*
4. *the accounting holds at the protocol's boundary over the extended support, and at H_f over the support the call's result binding and the restoration's renamings determine;*
5. *at the boundary the fresh cursor cell is empty, the cursor advances by one, both components are exclusive, and the boundary heap is heap-compatible with the abstract take-out heap; at the call's output heap the put-back selects `mutate`, writes the child into the parent's selected slot, and changes no other cell.*

```
theorem principal_counted_fragment
    (hproj : projectNested declName takeOut code = some p)
    (hl : NestedLayout assign p) (hm : LevelMatch assign p l path value rA o i
    childRef)
    (hag : NestedAgrees assign p s mEnv H rA i v pv)
    (hpath : PathContract path value body tip l rest H mEnv c cb y)
```

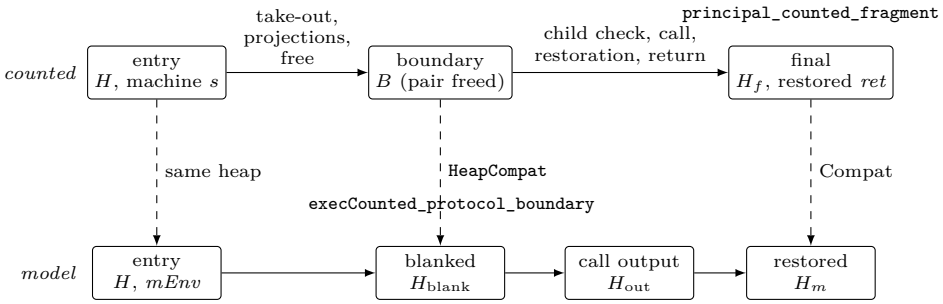


Figure 2 Simulation boundaries. The counted and model rows are connected only where a theorem relates them: at entry, after the pair's release (`execCounted_protocol_boundary`) and after restoration (`principal_counted_fragment`). Inside the pair-live window the accounting is not an invariant of the counted execution, so no vertical edge is drawn there; the suffix is run from the boundary state, and its restoration is derived at the call-output heap H_{out}.

```
(hacc : Ownership.Accounted support H mEnv c)
(hcall : Ownership.CountedChildCall
  (Ownership.CountedPair.protocol H rA childRef i).2 Hout ... ) :
∃ Hf envf ws Hm envm,
  execCountedAST ... = some ((execCounted ... p.fragment s).1, some (.done Hf
envf (.ref rA) ...)) ∧
  NestedOutcome.Compat (NestedOutcome.ofMachine ...) (NestedOutcome.ofModel
...) ∧
  Ownership.Accounted ... (Ownership.CountedPair.protocol H rA childRef i).2
... ∧ ...
```

Abbreviated: the full type, all conjuncts and the premise providers are in `OwnershipLcnf/Principal.lean`; Section 7.3 points at the declaration map.

Proof sketch.

The measure is the decomposition of the fragment, not a recursion on syntax. Recognition fixes operand identities and operation order, so the counted interpreter's fragment is a fixed sequence; `fragment_decomp` splits it at the release-terminated prefix `countedPrefix`—a syntactic split, not a numeric cutoff—giving `countedSuffix` as the rest. On the prefix, `execCounted_protocol_boundary` reports the protocol boundary heap, agrees with `execNested` on the non-heap fields including the write trace, and relates the two heaps by `HeapCompat`. On the suffix, `execCounted_full_fragment` runs the child assertion, the call at the boundary state, the restoration at the call's output heap and the return. The call is consumed where the interpreter reaches it, so the supplied contract is used at the actual call-entry state rather than at a reconstruction of it, and the restoration's mutate branch is derived at the actual output state. The two halves compose to `NestedOutcome.Compat`, and because the entry point is `execCountedAST`, the recogniser's verdict is spent by the theorem rather than assumed by its statement.

7 The instance, the case study and the artifact

7.1 A worked instance

The instance is a family. For an outer slot i, a leaf slot j and a value w, the entry holds two live, non-persistent cells at count one: the parent array at reference 0 carries the child handle

in slot i and scalars elsewhere, the child array at reference 1 is all scalars, and reference 2 is the fresh name. The child body is one designated write that stores w in the child's slot j and returns the child handle. Four declarations carry the family, for every $i < n$ and $j < n$: `entryAccounted` accounts the entry; `boundary_cells` and `boundary_accounted` report the protocol boundary and its accounting; `countedBoundaryCall` discharges the counted child-call contract; and `family_chain` states the whole `counted_nested_chain` conclusion at that data, with `family_chain_two` its length-two instance. The bound is the only length fact these proofs use: the slot facts are `Array.getElem?_setIfInBounds_self_of_lt`, and the count equation at the parent's and the child's own references comes from `IncomingCount_baseHeap`, `IncomingCount_set_fill` and `IncomingCount_set_fill_ne` on the all-scalar array.

Take $i = 0$, $j = 1$ and $w = 7$. Before the take-out the parent's slot 0 holds `.ref 1` and its slot 1 a scalar; after it, slot 0 holds the placeholder `.scalar 0`, the temporary pair has been allocated at name 2 and released, and the cursor stands at 3. The leaf then stores `.scalar 7` in the child's slot 1, and the restoration refills the parent's slot 0 with `.ref 1`. Three facts are separate and none implies the others: the restored array is functionally the model's own `set`; the counts are back to one at both components with the pair gone; and the restoration took the `.mutate` branch because the parent is exclusive at the call's output state, not because the counts happen to be unchanged. The final heap and the abstract run's heap are related by `HeapCompat`, not equality: the counted run's cursor is one higher, the residue of the temporary pair.

`principal_instance` inhabits `principal_supported_path` at the recogniser's layout on a reified transcription of the captured declaration, with every premise supplied, and `counted_fragment_instance` inhabits `principal_counted_fragment` at the same data; both are closed existentials whose `#print axioms` record is the trusted footprint. The instance discharges `CountedChildCall` through `calledAccounted`, which is the caller's own accounting at the body's return rather than an extra assumption: no theorem in the development derives that field from the path induction.

7.2 What the mechanisation taught

Four lessons transfer to anyone building another analysis of this kind. First, the tempting invariant—count equals multiplicity—is too weak, and the countermodel `alias_not_accounted` exposes it: the missing quantity is the environment's owning variables, not the heap's edges. Second, support and domain coverage are not bureaucracy: without the domain field the edge count is taken over a range that need not contain every live cell, so a cell's stored count and the enumerated edges can disagree. Third, the simple accounted relation cannot hold inside a correct temporary-alias protocol, and no amount of invariant engineering fixes that: the pair legitimately duplicates a reference before any count is incremented, so the right response is boundary simulation with internal count identities, not a stronger invariant that the protocol would have to satisfy. Fourth, allocator progress prevents plain equality, so correspondence between a counted run and an abstract one needs an explicit cursor slack; `HeapCompat` is that slack made a relation, and it degenerates to equality when the cursors agree. Finally, some restrictions follow from the chosen representation—the flat environment cannot express a re-projected index—while others are unimplemented features: general increments are excluded because no supported obligation carries them, whereas concurrency and persistence flags are simply outside the fragment.

Variation	What is observed	Which mechanism answers it
retained parent (count two)	the restoration copies; `.mutate` is not taken (`retained_parent_not_mutate`, `retained_parent_copies`)	the model's copy branch; the relation exposes the extra allocation through the cursor
pair not released	both counts elevated, no exclusivity (`no_free_components_not_exclusive`)	refused by the relation
pair-live state	no accounting at the take-out context (`pairPhase_refused`)	refused by the relation
duplicated, omitted or unbound owner entry	each field is load-bearing: a duplicate double-counts a root, an omitted owner still satisfies the equation, an unbound entry counts nothing (`dupSupport_witness`, `omittedOwner_witness`, `unboundRoot_witness`)	refused by the relation
persistent parent	the check aborts, no write is reported (`persistent_check`, `persistent_setAt`)	rejected by the checker; outside the certified fragment

Table 3 Controls. The mechanisms are not interchangeable: a row refused by the relation is not a checker rejection, and a copy branch in the model is not an allocation claim about the executable.

7.3 Evidence and artifact

The proof checks are kernel builds of `Ownership`, `OwnershipTest` and `OwnershipLcnf`, an axiom audit over the headline theorems, and the artifact gate that binds the pinned LCNF, C and header. They are separate categories from the structural recogniser controls (the near-miss counts and the control fixtures of Table 3) and from the linked runtime observations. A compiled probe compares the model's predicates with `lean_is_exclusive` and address identity on a corpus including a unique array, a live alias, the alias dropped back to one, a persistent literal and a closure-held array; its differential mode compares recorded expectations rather than executing the `Ownership` model, and its closed-pure-array case is an unchecked observation recorded in `docs/notes/OWNERSHIP_RUNTIME_EVIDENCE.md`. The linked gate observes the primitives' behaviour on the certified shapes, including whether the write changes the address, and proves none of it. The declarations behind every claim, with their roles, are listed in the supplementary document `docs/notes/ownership-theorem-map.md`, which is artifact documentation and not part of the argument; the paper-to-code entry points are `Ownership/CompleteAccounting.lean`, `Ownership/Preservation.lean`, `Ownership/CountedNested.lean`, `OwnershipLcnf/CountedTrace.lean` and `OwnershipLcnf/Principal.lean`.

8 Related work

Huisinga's 2023 thesis [4] is the closest predecessor and the comparison is exact. It analyses a model of Lean's IR, enforces uniqueness through a static analysis, and studies that analysis's checker; uniqueness there is a property of the analysis, not of the runtime count; it has no compiler link; and its soundness argument and compiler integration are stated as future work. The difference is not that our proofs are machine checked—theirs could be too—but what is connected: here a count-one checker is proved against a model heap with an explicit roots-and-edges accounting, that result is composed with a transform of the recognised

Category	What it covers	Status
checked syntax	the recogniser's accepted grammar; the two model languages	checked by elaborating `#guard` tests and by the recogniser itself
proved model facts	accounting and preservation; protocol boundaries; counted fragment run; the runtime model's in-place statements	theorems in `Ownership` and `OwnershipLcnf`, no axioms beyond Lean's
supplied client and call facts	entry invariant, agreement, layout and path contract, the leaf computation, and `CountedChildCall`'s `accounted` field	hypotheses of `principal_supported_path` and `principal_counted_fragment`; discharged at the closed instance
implementation assumptions	primitive fidelity, replay-to-backend identity, reference identity	assumed or empirically observed; not proved

Table 4 The trust boundary, by category.

emitted prefix and with a second runtime model at the flat write, and the extraction protocol is isolated at proved accounted boundaries. This is not a claim to the first static uniqueness analysis for Lean.

Perceus [9] and Counting Immutable Beans [10] are count-insertion and reuse transformations: an ownership pass inserts counts and reuse tokens, the runtime maintains them, and the correctness argument is about the transformation's semantics preservation and its allocation behaviour. They do not answer the question asked here, which is when a count of one is justified at a given write in code that already exists; `Accounted` is a specification of that justification, and this paper proves it for a restricted fragment rather than proving a transformation. Linear Haskell [2] makes linearity a source-level type property through linear arrows and multiplicities, and Clean's uniqueness typing [1] is uniqueness rather than linearity; neither is a checker over an existing compiler IR, and neither is connected to emitted code. Lean's `Array.markLinear` [6] is dynamic: it can copy an array to establish uniqueness and, with `LEAN_ABORT_ON_NONLINEAR`, abort on a later nonlinear use. Translation validation [8] checks one compiler run rather than a static discipline, and verified compilation [7, 5] proves a compiler correct for its source language; both are different in kind from a checker over a fragment of an existing backend's input.

9 Scope, conclusion and disclosure

▷ Non-claim 1. The C semantics of the primitives, replay-to-backend identity, address identity, free and cascade, and the source correspondence are neither formalised nor attempted. The emitted recursive callee and continuation (N10) are not captured, and the flat environment cannot express the per-invocation index, so a deeper level that re-projects its index is outside the model. The leaf body and several heap-shape facts are supplied. Persistence, concurrency and callbacks are outside the fragment, and the nested linked runtime is not measured.

The accounting layer is not a consequence of `Inv`: `OwnersAccounted` has a kernel counter-model, and the checker's certificates are carried by `Supported`, whose fragment excludes `.inc` and admits only `.dec x 1`. `StoreSlots` is supplied, and `storeSlots_init` covers only the initializer's measured write. `principal_write` takes no accounting premise and concludes

none; `principal_supported_path` takes the caller's accounting and the counted child call and concludes accounting at the counted boundary and at the restored state; no statement carries either relation to a physical counted reference, and no step-by-step agreement between the two interpreters inside the pair-live window is claimed. The nested trace omits extraction and restoration stores—neither interpreter records them as `Write` records—but each target cell is proved exclusive (`checkN_extraction_exclusive_of_entryInv`), a fact about the store's target rather than a trace event, and not a silent absence. Whole-path non-growth outside the root's mutable footprint is not concluded; it needs the leaf's write account. "In place" throughout is the modelled disposition: the array operation neither copies nor allocates under the contracts these statements take, and it is not a zero-allocation claim for the protocol, whose take-out allocates the pair. The pair-arm correspondence with `Ownership.Pair.runP` is a named next step.

Conclusion.

A count-one discipline can be recovered for an existing compiler IR without changing the language, provided the accounting is complete rather than local, and provided the emitted protocol is treated as a boundary phenomenon rather than an invariant. That is what this paper establishes: a proved relation between owning roots and incoming edges with the uniqueness consequence that makes a stored count one meaningful; preservation for a restricted flat fragment, with the slot fact isolated as the one premise the checker cannot see; a counted extraction, call and restoration protocol proved at its accounted boundaries, with the in-place restoration branch derived at the call's output state; and, for the full counted fragment at one level of a finite supported path, a checked correspondence to the model's run—heap-compatible, with the same result, fault and write trace—under an explicit counted child-call contract that the closed instance discharges. The result buys a checked count one for that fragment, not a type system and not a physical count. Next come the C semantics, replay and address identity, free and cascade, and the emitted callee. Soundness is proved; completeness is not claimed.

Acknowledgements

AI assistance.

The development and this manuscript were produced with substantial AI assistance. Tools and model versions used across the development were not recorded. The model identifier `deepseek/deepseek-flash` is recorded for this manuscript revision in `docs/notes/artifacts/final-revision-check.txt`. Categories of assistance: prose, Lean proofs, code, analysis and literature search. The division of labour between the author and these tools is not recorded. Outputs were checked by the pinned kernel build and the axiom audit, and a declaration census checks each displayed name against the development; no line-by-line human review of the proof scripts is recorded. Kernel checking is not human validation: the definitions, theorem statements, interpretation of evidence, novelty claims and citations are the author's responsibility. The conference and publisher policy must be rechecked at submission.

References

1 Erik Barendsen and Sjaak Smetsers. Uniqueness typing for functional languages with graph rewriting semantics. *Mathematical Structures in Computer Science*, 6(6):579–612, 1996.

2 Jean-Philippe Bernardy, Mathieu Boespflug, Ryan R. Newton, Simon Peyton Jones, and Arnaud Spiwack. Linear haskell: Practical linearity in a higher-order polymorphic language. In *Principles of Programming Languages (POPL)*, 2018.

3 Leonardo de Moura and Sebastian Ullrich. The lean 4 theorem prover and programming language. In *Automated Deduction – CADE-28*, volume 12699 of *LNCS*. Springer, 2021.

4 Marc Huisinga. Static uniqueness analysis for the lean 4 theorem prover. Master's thesis, Karlsruhe Institute of Technology, 2023. URL: `https://pp.ipd.kit.edu/uploads/publikationen/huisinga23masterarbeit.pdf`.

5 Ramana Kumar, Magnus O. Myreen, Michael Norrish, and Scott Owens. Cakeml: A verified implementation of ml. In *Principles of Programming Languages (POPL)*, 2014.

6 Lean FRO. Array.marklinear, 2025. Lean API documentation. URL: `https://lean-lang.org/doc/api/Init/Data/Array/Basic.html#Array.markLinear`.

7 Xavier Leroy. Formal verification of a realistic compiler. *Communications of the ACM*, 52(7), 2009.

8 George C. Necula. Translation validation for an optimizing compiler. In *Programming Language Design and Implementation (PLDI)*, 2000.

9 Alex Reinking, Ningning Xie, Leonardo de Moura, and Daan Leijen. Perceus: Garbage free reference counting with reuse. In *Programming Language Design and Implementation (PLDI)*, 2021.

10 Sebastian Ullrich and Leonardo de Moura. Counting immutable beans: Reference counting optimized for purely functional programming. In *Implementation and Application of Functional Languages (IFL)*, 2019.